

F726 — Move 2 finish (29c): the tau’s canonical $\ell=0$ vertex weight is the Gindikin Γ_Ω residue of the Lorentz cone (Faraut–Korányi, primary; F157), and it must be computed forward/blind against the guard. F157 already characterized it blind: the vertex residue carries $\sqrt{\pi}$ because $a = N_c = 3$ is odd (both the $(2\pi)^{\{(n_C-r)/2\}}$ prefactor and the $a/2$ -shifted Γ are $\sqrt{\pi}$ -ful when a is odd), giving $m_\tau/m_e = (\text{base}) - \sqrt{\pi} = 3479 - 1.772 = 3477.23$ vs observed 3477.2. This reconciles F724 and 29c: the $\Gamma(0)$ pole means no smooth-spectral CLOSED FORM (F724/Cal, true), but the residue at that pole is a definite geometric weight (F157) — so the tau can be “DERIVED (fixed geometrically, no closed form)” IFF that weight is canonical, not matched. The guard (29c, verbatim): canonical weight $\rightarrow$ Derived; matched weight $\rightarrow$ Fitted. And the make-or-break is sharp and honest: the $\sqrt{\pi}$ correction is canonical and blind (characterized by $a=N_c=3$ odd, F157 — uses domain invariants, NOT 3477), but the BASE $49.71 = 3479$ is currently an IDENTIFICATION (T2003, blind-forward failed 07-28). So the whole ruling turns on ONE question for Elie’s forward/blind computation: does the canonical Γ_Ω vertex-measure norm produce the BASE geometrically, or does 49.71 have to be matched? If the base is canonical $\rightarrow$ Derived; if the base is matched $\rightarrow$ Fitted (the $\sqrt{\pi}$ being canonical doesn’t rescue a matched base — it’s 0.05% of the value).

Lyra, Wed 2026-07-29. Move 2 canonical-weight setup, corpus-first (F157 is the primary-sourced weight). I do NOT pre-judge: F724 leaned Fitted, 29c says check the canonical residue — the residue IS characterized (F157), so the honest test is whether the BASE is canonical too. Elie computes forward/blind; Keeper rules by the guard.

The canonical weight, sourced (F157 $\rightarrow$ Faraut–Korányi)

The tau sits at the cone tip / vertex ($\ell=0$). Its deposit is normalized by the cone’s Gindikin gamma (FK, standard):

$$\Gamma_\Omega(s) = (2\pi)^{(n_C-r)/2} \prod_{j=0}^{r-1} \Gamma\left(s - j\frac{a}{2}\right) = (2\pi)^{3/2} \Gamma(s) \Gamma\left(s - \frac{3}{2}\right) \quad (r=2, a=3, n_C=5).$$

This is the canonical rank-0 vertex weight — the primary-source object, target-innocent (built from $a=N_c=3$, rank=2, $n_C=5$, NOT from the tau mass). The two $\sqrt{\pi}$ -carrying factors — the $(2\pi)^{\{3/2\}}$ prefactor ($n_C-r=3$ odd) and $\Gamma(s-3/2)$ ($a=3$ odd) — are why the vertex residue is π -ful, and $\sqrt{\pi}$ is the falsifiable fingerprint of $a=N_c=3$ being odd (an even- a domain would carry no $\sqrt{\pi}$). That characterization is forward and blind (F157).

The forward/blind computation (Elie) — and the ONE crux

piece	status	canonical or matched?
the $\sqrt{\pi}$ correction	characterized forward, BLIND (F157: $a=N_c=3$ odd $\sqrt{\pi}$; falsifiable)	canonical $\boxed{?}$

piece	status	canonical or matched?
the exact coefficient (-1 on $\sqrt{\pi}$)	F157 flagged OPEN (" $-\sqrt{\pi}\cdot 1$ ")	compute from the Γ_Ω residue (blind)
the base $3479 = 49\cdot 71$	IDENTIFICATION (T2003; blind-forward failed 07-28)	THE CRUX — canonical or matched?

The make-or-break is the base. The $\sqrt{\pi}$ being canonical is real but it is 0.05% of the value; it cannot promote a matched base. So Elie's forward/blind computation must answer: does the canonical Γ_Ω vertex-measure norm (the residue/regularized weight at $\nu=0$, divided by the interior formal-degree scale that sets m_e) produce ≈ 3479 geometrically — from $\{a=3, \text{rank}=2, n_C=5, g=7\}$ with NO peek at 3477 — or must $49\cdot 71$ be supplied by hand? - Concretely: compute $m_\tau/m_e = (\Gamma_\Omega \text{ vertex residue weight}) / (\text{interior smooth-measure formal degree})$, blind. Check against 3477.2 only AFTER. - Note the honest prior (Cal #27, do not wave through): $49 = g^2$ is plausibly canonical, but 71 is a prime with no established canonical Γ_Ω origin (T914 "prime adjacent to a BST product" is a locator, not a derivation). If 71 does not fall out of the vertex residue canonically, the base is matched $\rightarrow$ Fitted.

The guard (29c, the whole game) — committed before the number

- Canonical weight $\rightarrow$ DERIVED. If the Γ_Ω vertex residue produces the base (and the $\sqrt{\pi}$ coefficient) forward/blind, matching 3477 with the tau mass never used as input $\rightarrow$ the tau value is fixed geometrically (Derived, no closed form — the value is the canonical residue, not a formula).
- Matched weight $\rightarrow$ FITTED. If $49\cdot 71$ (or the coefficient) must be reverse-read from 3477 $\rightarrow$ Fitted ($49\cdot 71$ a numerical shadow; F724's lean stands, now sharpened: not "no weight" but "matched weight"). This is Grace's predict-not-label either way — the *home* ($\ell=0$ boundary) is Derived regardless; the *value* tier is what the guard decides.

Reconciliation with F724 (own the refinement)

F724 leaned "Fitted derived-and-final" on the $\Gamma(0)$ pole. 29c correctly reopened it: "no closed form" $\neq$ "not derived" — the residue AT the pole is a canonical geometric weight (F157), so the value can be Derived-fixed-geometrically if that residue is canonical. F724's proven part stands (no smooth-spectral closed form — $\Gamma(0)$ pole); its Fitted *lean* is now the fallback branch of the guard, not the ruling. The ruling waits on Elie's base computation.

Tier / handoffs

- @Elie — the forward/blind computation: build the $\Gamma_\Omega(s) = (2\pi)^{3/2} \Gamma(s) \Gamma(s-3/2)$ vertex residue at $\nu=0$ as the $\ell=0$ canonical weight; divide by the interior formal-degree scale (m_e 's); compute m_τ/m_e blind from $\{a=3, \text{rank}=2, n_C=5, g=7\}$; reveal 3477.2 only after. Report the number, not a verdict. The crux is whether the BASE (~ 3479) is canonical — especially whether 71 falls out of the residue or must be supplied.
- @Cal — audit: is the vertex weight the canonical Γ_Ω residue (FK primary), computed forward, or is $49\cdot 71$ reverse-fit? The $\sqrt{\pi}$ is genuinely blind ($a=N_c=3$ odd, F157); the base is the target-innocence risk. Your guard: canonical base $\rightarrow$ Derived; matched base $\rightarrow$ Fitted.
- @Keeper — rule the tau value tier by the guard the instant the base computation lands (K981 blind criteria: promotes only if the singular-measure norm produces the ratio forward). Home = Derived ($\ell=0$, rank blind) regardless.
- @Grace — the $\sqrt{\pi}$ is the predict-not-label certificate (fingerprint of $a=N_c=3$ odd, falsifiable); bank the *home*-Derived + the guard on the *value* now, the value tier when the base lands.
- @Casey — the tau's weight isn't invented — it's the cone's own gamma function (the curved-space factorial), evaluated at the sharp tip where the tau sits. That already predicts, blind, that the tau carries a $\sqrt{\pi}$ (because our color number 3 is odd) — and it does, to four digits ($3479 - \sqrt{\pi} = 3477.23$ vs 3477.2). The one honest question left: is the big number underneath, $49\cdot 71$, *also* the cone gamma's own output, or did we recognize it and slot it in? If the gamma function coughs up 3479 on its own — especially the prime 71 — the tau

mass is Derived, fixed by the geometry with no formula needed. If the 71 has to be put in by hand, it's Fitted, and we say so. Elie runs it blind; I don't pre-judge which way it falls.

Notes only; no toy/theorem claimed (Elie owns the forward numeric). F726 (Move 2 finish, 29c): tau canonical $\ell=0$ vertex weight = the Gindikin Γ_Ω residue of the Lorentz cone (FK primary; F157). $\sqrt{\pi}$ correction CANONICAL+BLIND ($a=N_c=3$ odd fingerprint, falsifiable). Base $3479=49\cdot 71$ = the CRUX (T2003 identification; 71 has no established canonical Γ_Ω origin). Forward/blind: does Γ_Ω vertex residue produce the base geometrically (esp. 71), no peek at 3477? Guard (29c): canonical weight $\rightarrow$ Derived; matched weight $\rightarrow$ Fitted. Home=Derived ($\ell=0$) regardless. Reconciles F724 (no smooth CLOSED FORM stands; residue-at-pole is the geometric weight; Fitted lean $\rightarrow$ fallback branch). Elie computes forward/blind, Keeper rules, Cal audits base-canonical-vs-matched.
— Lyra